

		params	L1 known rule			L2 seen worlds			L3 unseen worlds			L4 with prior		
Standard models														
	CNN	116k	100	100	100	22.7	22.5	22.6	19.0	18.8	19.0	11.3	7.8	10.9
	transformer, 2 layers (d64)	166k	100	100	100	21.0	31.6	33.7	17.8	28.6	31.2	3.3	5.8	6.7
	transformer, 4 layers (d128)	925k	100	100	100	91.6	79.0	89.6	92.6	76.4	87.8	82.2	65.3	88.4
	transformer, 6 layers (d192)	2.87M	100	100	100	50.9	5.4	62.1	49.0	4.1	58.9	67.7	0.2	61.5
	transformer, 8 layers (d256)	6.58M	100	100	100	5.4	86.2	70.6	4.1	84.7	69.2	0.2	0.2	0.2
	diffusion	86k	19.1	28.1	27.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Spatial locality (Sec. 3)														
	transformer, raster RoPE, 2 layers (d64)	100k	39.2	39.6	38.6	19.5	20.7	20.8	16.6	17.4	17.5	1.3	1.4	1.3
	transformer, axial RoPE, 2 layers (d64)	100k	100	100	100	19.2	19.2	19.3	16.3	15.5	15.7	0.6	0.9	1.3
Temporal locality (Sec. 4)														
	transformer, 2 layers (d64) + last-frame 3×3	167k	100	100	100	99.9	99.9	99.9	99.8	99.9	99.9	66.2	59.3	94.1
	transformer, 4 layers (d128) + last-frame 3×3	926k	100	100	100	99.9	99.9	99.9	99.9	100	99.9	98.8	100	92.8
	2D-conv transformer, 2 layers	142k	100	100	100	95.8	91.1	80.5	96.1	90.9	79.5	84.3	81.7	73.8
	KV-shift transformer, 1 memory read	2.6k	100	100	100	100	100	100	100	100	100	16.1	51.8	58.7
	KV-shift transformer, 4 memory reads*	9.8k	100	100	100	100	100	100	100	100	100	98.8	99.8	99.8
	3D-conv transformer, 1 memory read	3.8k	100	100	100	100	100	100	100	100	100	71.3	99.8	43.3
	3D-conv transformer, 4 memory reads	11k	100	100	100	100	100	100	100	100	100	100	99.8	98.7
	3D-conv CNN	18k	100	100	100	100	100	99.4	99.8	100	99.5	100	100	96.6
Temporal stability (Sec. 5)														
	per-frame noise + backward encoder + anchor	30k	31.8	32.0	32.2	6.2	6.2	6.1	4.5	4.7	4.3	0.0	0.0	0.3
	same denoiser, frame-ordered sampling	30k	100	100	100	25.1	33.4	34.1	21.0	30.1	32.8	36.8	9.5	28.2
	same denoiser, order-matched training	30k	100	100	100	100	99.1	98.8	100	99.2	98.9	100	84.1	95.4